1. Core Python.
2. Database with Python.
3. API (flask)
4. Scrapping
5. $\boxed{\text{Stats.}}$
6. EDA (Pandas, numpy, matplot | seaborn)
7. ML
8. End to end Proj
9. basic of DL

STATS ⇒ mlBootcamp.

1. Data.
2. Descriptive stats
3. Inferential stats

Descriptive stats →
- Central tendency of the data
- Dispersion method.
- Probability theory
- Probability dist. function

Inferential stats →
- Sample | Population
- hypothesis testing
- Z-test
- t-test
- f-test
- anova test
- Chi-square test

Stats ⇒ Part of the math.

Def. ⇒ Collect, organizing, analysis

Data ⇒ facts or piece of information

eg ⇒ IQ of student

Height of the student in classroom

Daily activity

weight, age of the people

types of stats

Descriptive stats

Def — Summarizing our data with the single value

one value.

inferential stats

Def find out the conclusion for the popula- -tion based on sample data.

City ⇒ People. ⇒ height

City → Sample $\boxed{\text{hypothesis}}$ testing.

Population

- CI, P, Sig. level, one tail two tail test, standard error

[z-test, t, f, Anova, chi-square.]

Data

- Structure data
  - Data which is av. in RxC (table)
  - Numeric data
    - Continuous
      - weight, height, temp.
      - 5.5
      - 6.1
      - 6.22
    - Discreate
      - whole number
      - no of student in the days
      - ~~50.5~~ ~~60.5~~
      - 50
      - 60
      - 55
  - Categorical data
    - Nominal
      - there is not any order wrt category
      - Gender
      - blood grp
      - Color of Hair
    - Ordinal
      - Rank
        - Best
        - Better
        - good
      - Education.
        - 10
        - 12
        - BE
        - mtech
        - Phd
      - Order
- Unstructure data
  - text, video, image, voice,
  - (Json) (DL)

Data tendency

- Batch data
- mini batch data
- Streaming data

Class.

- male
- female

Blood.

A, AB, B+, B-

~~Order~~

White, black, mild black, Red, brown.

~~Order~~

Data ⇒ Student Performance ⇒ analysis → You know the stats.

Dataset ⇒ structure dataset (RXC) ⇒ each col is feature/attribute

Roll no.	Name	Degree	Percentage	Height	Weight	Gender
1	Amit	BE	85	5.5	60	M
2	Sunny	Btech	80	5.3	65	M
3	Suresh	Phd	75	6.1	62.2	M
4	Komal	Mtech	82.8	5.2	50.2	F
5	Ashi	MBA	95.5	5.7	55.5	F

Name ⇓ Cat — nominal.

Degree ⇓ Cat ⇒ Ordinal. Order ✓

Percentage ⇓ Num ⇓ Cont.

Height: Num → Cont.

Gender: Cat ⇒ nominal. ~~Order~~

Name	description	sentiment
Sunny	Sunny is DS who is working in ineuron good experience ML and DL he is living in BLR	Positive.

text data

NLP approach (pattern) text pre NLP

Descriptive stats

1. mean.
2. median
3. mode

Data

class room.

students

all data { 5.5, 6.2, 5.1, 5.3 .... }

avg height (mean)

5.7

= one single value

mean ⇒ { 1, 1, 2, 2, 3, 3, 4, 5, 5, 6 }

describe your data with one single value = ?

$$\frac{\sum_{i=1}^{n} x_i}{N}$$

$$\Rightarrow \frac{1+1+2+2+3+3+4+5+5+6}{10} = 3.2$$

Population
Sample } inferential stats (Notation)

mean → [10, 20, 30, 40, 50, 60, 70] ⟹ [10, 40, 50, 60]

Population

sample

Measure	Population Parameter	Sample Statistic
Mean	μ	$\overline{X}$
Variance	σ^2	S^2
Standard Deviation	σ	S

notation.

Data ⇒ [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

mean. ⇒ μ

Sample mean. ⇒ [10, 50, 60, 90]

$\overline{X}$

mean ⇒ 1, 2, 3, 4, 5

$\Rightarrow \frac{15}{5} = 3$

1, 2, 3, 4, 10

$\Rightarrow \frac{20}{5} = 4$

median. ⇒

middle value.

1, 2, $\boxed{3}$, 4, 5 ⇒ Sorted.

1, 2, $\boxed{3, 4}$ 5, 6

$\Downarrow$

$\frac{3+4}{2} = 3.5$

Q. 1, 5, 4, $\boxed{3}$, 10, 7, 8

median

$\Downarrow$

sort

1, 3, 4, $\boxed{5}$ 7, 8, 10

avg = 3ft
height of the person = 5ft

1 2 3 4 5

$\Rightarrow$ mean = 3

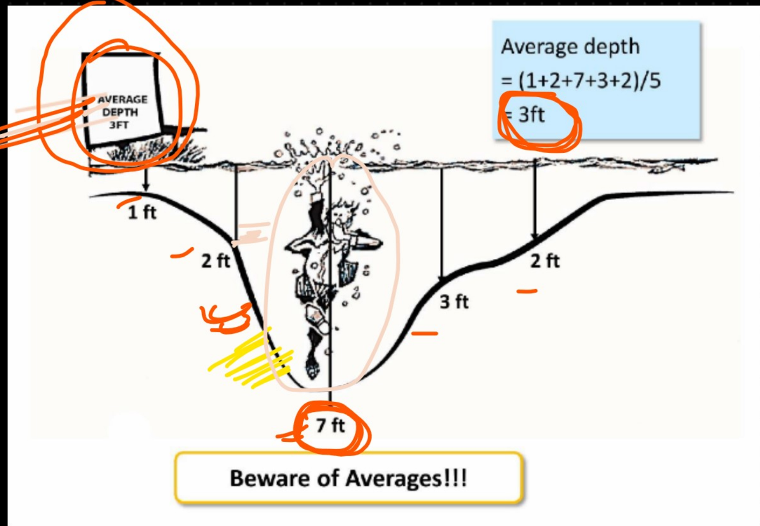
median $\Rightarrow$ 3

1, 2, 3, 4, $\boxed{10}$

Outliers

$\Rightarrow$ mean. 4

median 3



1, 2, 2, 3, $\boxed{7}$

1

Pond/river

3-ft
person-5ft

mean ⇒ avg

median → Odd

⇒ 1, 2, $\boxed{3}$, 4, 5 → 5

⇒ 1, 2, $\boxed{3, 4}$, 5, 6 ⇒ 6

⇓

$\frac{3+4}{2} = 3.5$

Deal #	Deal Value	Deal Status
1	70,000	Open
2	50,000	Closed
3	55,000	Closed
4	60,000	Closed
5	55,000	Closed
6	50,000	Closed
7	50,000	Closed
8	60,000	Closed
9	50,000	Closed
10	5,00,000	Open

1, 2, $\boxed{2}$, 3, 3, 7

↑

min max

avg / mean

spoil

balance

1, 2, 3, 4, 5, 60, 70, 80, 90, 100

⇑

median

Mode ⇒

1, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 5, 6

⇓

5 most frequent value in the dataset

Categorical.

Categorical data

unsort

⇒ 1, 12, 13, 6, 5, 12, 5, 3, 2, 5, 6, 9

⇒ mode ⇒ ? freq. of the data element

most freq. value

first sort

1, 2, 3, $\boxed{5}$, $\boxed{5}$, $\boxed{5}$, 6, 6, 9, 12, 12, 13

Descriptive stats ① central tendancy

② Dispersion of the data

Range
Percentage, percentile, quartile [5 no summray of the data]
variance, (cov variance, corr)
Std dev

(relationship b/w two variable)

③ Probability theory,
Random var,
pdf

1, 2, 3, 4, 5, 6, 7

Range
Percentile
variance → mean
Std = $\sqrt{var}$